SIR

$S \rightarrow$ Susceptibles
$I \longrightarrow$ Infectives
$R \longrightarrow$ Removed / recovered

assumption:

population remains constant
Rate of infection $\propto$ contacts but person
Infective recover/die at constant rate

$$\frac{dS}{dt} = -\gamma I S$$

$I \rightarrow$ infective (infected)
$S \rightarrow$ susceptible to infection
$\gamma \rightarrow$ contact probability
$a \rightarrow$ Recovery ~~rate~~ probability

$$\frac{dI}{dt} = \gamma I S - a I$$

$$\frac{dR}{dt} = a I$$

initially $S = S_0, \ I = I_0, \ R = 0$

And $\dfrac{d(S+R+I)}{dt} = 0$

$\Rightarrow$ $S + R + I_0 = S_0 + I_0$

Q1 who will spread the disease?

$$S \leq S_0$$

$$\frac{dI}{dt} \leq I(\gamma S_0 - a)$$

for infection to occur $\gamma S_0 - a > 0$

$$S_0 > \frac{a}{\gamma} \qquad \frac{a}{\gamma} = \frac{1}{q}$$

$$S_0 > \frac{1}{q}$$

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On rearranging we get Parameter R_0

$$R_0 = \frac{\gamma S_0}{a} > 1$$

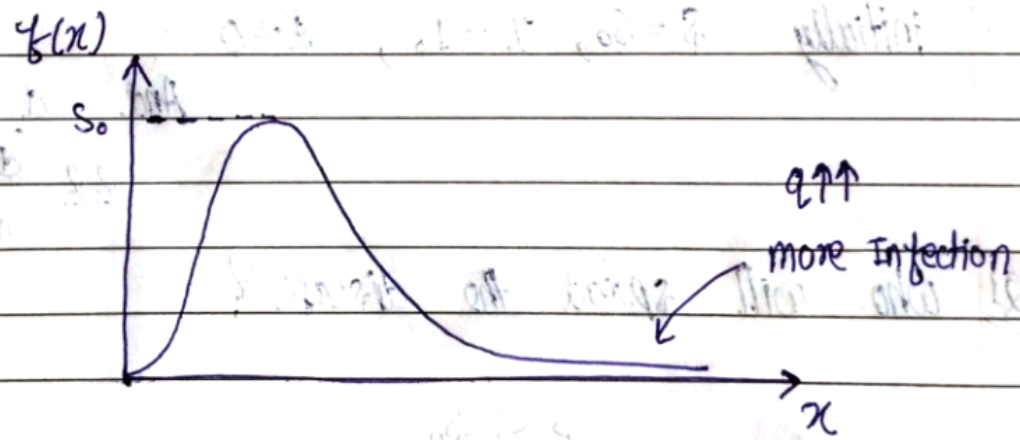
Q2 What will be maximum number of infectives?

divide eq (1) and (2)

$$\frac{dI}{dS} = \frac{\gamma I S - aI}{-\gamma I S} = -1 + \frac{1}{qS}$$

$$I + S - \frac{1}{q}\ln S = I_0 + S_0 - \frac{1}{q}\ln S_0$$ $\left(\frac{dI}{dS} = 0 \text{ then } q = \frac{1}{S}\right)$

$$I_{max} = I_0 + S_0 \underbrace{- \frac{1}{q}\left(1 + \ln(qS_0)\right)}_{f(x)}$$



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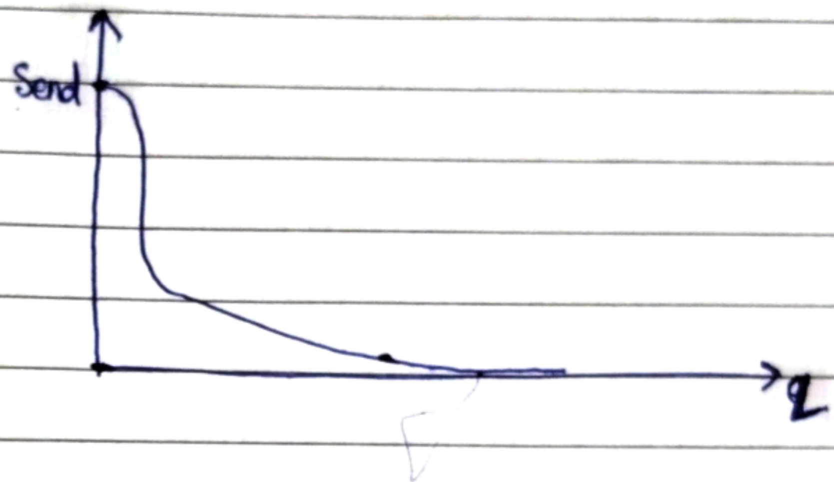
Q3 How many people caught infection?

$$R_{end} = -S_{end} + I_0 + S_0$$

Recovered people

$$S_{end} - \frac{1}{q}\ln(S_{end}) = I_0 + S_0 - \frac{1}{q}\ln(S_0)$$



- Spread $\Rightarrow$ $R_0 = qS_0 > 1$
- I_{max} = total population $- f(q)$
- Total = total population $- g(q)$